IIIT PUNE / HUMAN–COMPUTER INTERACTION + MACHINE LEARNING

# SignBridge

## Real-Time Indian Sign Language Recognition in the Browser

A user-centred HCI study and on-device ML prototype

Bittu Mandal (112415048) · Chinmay Ghule (112415054) · Dev Krishan (112415057) · Himanshu Raj (112415082) — IIIT Pune

## 01 PROBLEM

- Few hearing people know ISL. Names, numbers and spellings are hard to share without an interpreter.
- Accuracy alone misses how results are shown and how users recover when the system is wrong.

## 02 USER GROUPS

**PRIMARY**

ISL users · Hearing communication partners: family, staff and classmates

**SECONDARY** ISL learners · Technical evaluators

## 03 RESEARCH METHOD

10-question survey · 10 respondents · 6 literature sources

User survey → Findings → 3 personas → Literature review → Research gaps → Wire-frames → Working prototype → Testing

Survey → design requirements → tested prototype

## 04 PERSONAS

Survey-derived composites; names and ages are illustrative.

**Meera, 24**
Deaf ISL signer
Names and numbers without an interpreter; apps need reliability.

**Rohit, 21**
Hearing classmate; no ISL
Real-time text + speech, with a way to fix wrong results.

**Sana, 20**
ISL learner
Speech, privacy, and typing when a sign is not recognised.

## 05 KEY FINDINGS

SURVEY · n = 10

- **8/10** rate communication very or extremely difficult.
- **7/10 want speech; 7/10 want text** (5 want both).
- **Correction** is the most-wanted feature (3/10).
- When unsure: **4 ask to repeat; 3 want to type.**
  Show uncertainty and offer a recovery path.
- Mean usefulness: **3.0/5.** Reliability must be earned.

## 06 LITERATURE INSIGHTS

- **Design beyond accuracy.** Involve Deaf users.
  Bragg et al., ASSETS 2019
- **Word signs need temporal modelling.** INCLUDE uses video.
  Sridhar et al., ACM MM 2020
- **Static letters suit hand keypoints.** On-device tracking enables a lightweight pipeline.
  Zhang et al. 2020; Velmathi & Goyal 2023

## 07 RESEARCH GAPS

- Results need confidence and a clear correction path.
- Server recognition adds privacy and internet barriers.
- Word-level video models exceed this prototype's scope.
- Controlled datasets and frame leakage inflate accuracy.

## 08 SYSTEM

Camera (User-controlled) → MediaPipe (2 hands × 21 points) → Features (85 normalised) → MLP (85-128-64-35) → Smoothing (12 frames; ≥60%) → Output (Result + speech)

Locks when 9 of 12 frames agree and confidence ≥60%.

**Runs 100% in the browser — no video leaves the device**

## 09 ML RESULTS

**97.86%**
MLP · BLOCKED TEST ACCURACY

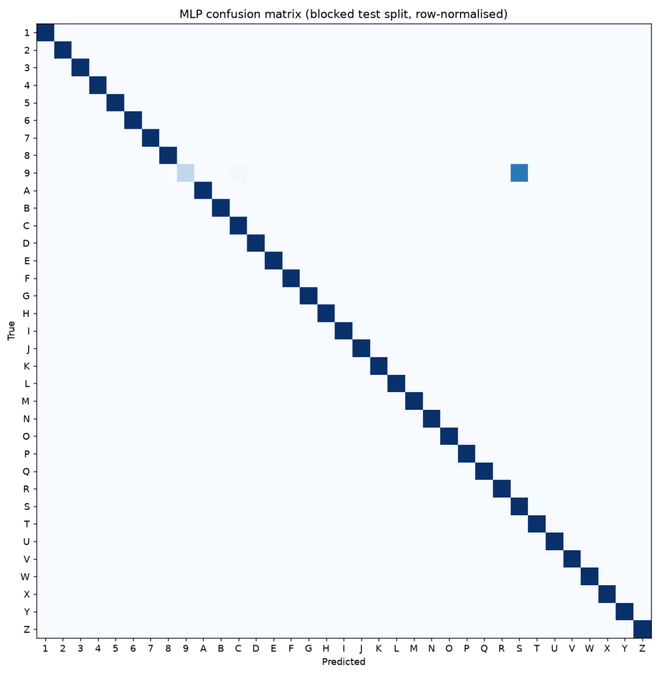
**35 classes**
STATIC FINGERSPELLING · A–Z, 1–9

42,745 source images · Blocked test: 8,377 images

| Model | Accuracy |
|---|---|
| kNN | 97.79% |
| Random Forest | 97.88% |
| SVM | 97.88% |
| **MLP (chosen)** | **97.86%** |

**Chosen for deployment**
0.46 MB model; inference in plain TypeScript.

**Split matters**
Random split: **99.95%**.
Near-duplicate frames inflate the result.

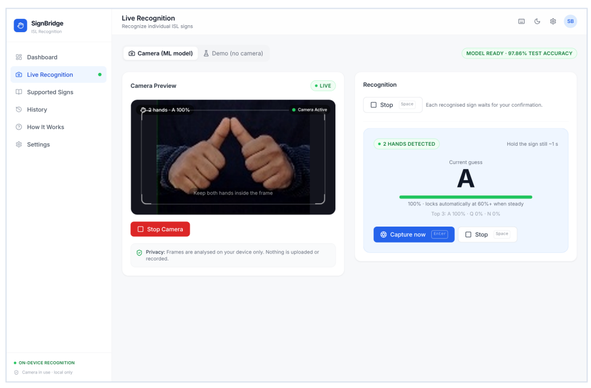


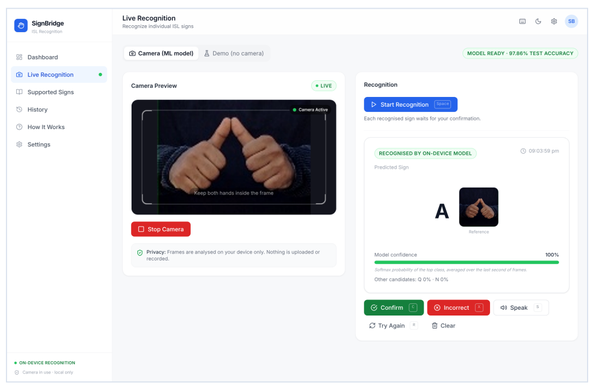

**Main error: 9 → S**
172 held-out frames confused.

Last 20% of frames per class held out. Generalisation to new signers and backgrounds still needs live evaluation.

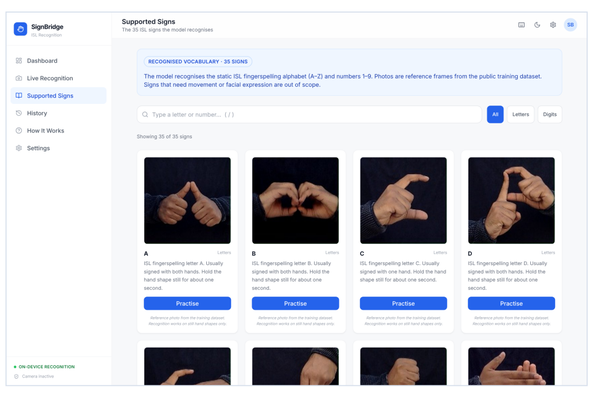
## 10 PROTOTYPE

Original interface captures and low-fidelity design



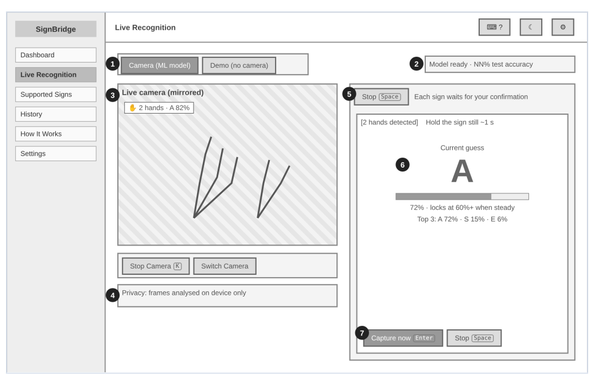




Live recognition with hand skeleton and live confidence




Supported signs with reference photos

Result with reference photo, confirm / incorrect / undo




Low-fidelity wireframe (W2)

UI demonstrations use a recorded test feed.
They are not a live-user evaluation.

## 11 SHNEIDERMAN'S 8 GOLDEN RULES

- **Consistency** · Same layout and colours
- **Shortcuts** · Space, C, X, S, R, U, ?
- **Informative feedback** · Hand skeleton + live confidence
- **Closure** · "Saved" notice + View history
- **Error prevention** · 60% threshold; uncertain state
- **Easy reversal** · Undo everywhere
- **User control** · Camera + recognition on command; auto-save toggle
- **Low memory load** · Reference photo beside every result

## 12 CONCLUSION & FUTURE WORK

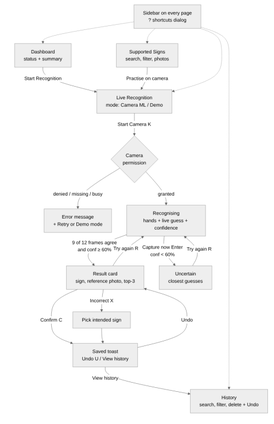
Real-time, private ISL fingerspelling recognition, with an interface built around uncertainty and correction.

**NEXT**

**Data** · More signers and backgrounds

**Models** · Word-level signs (sequence models)

**Evaluation** · Study with Deaf ISL users




Interaction flow

**REFERENCES** Bragg et al. 2019; Adeyanju et al. 2021; Sridhar et al. 2020; Velmathi & Goyal 2023; Zhang et al. 2020; Shneiderman et al. 2016.